

# Joonsuk Bae

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*Experimental high-energy nuclear physicist building methodological infrastructure for precision hard-probe measurements. Main analyzer of the first ALICE Run 3 inclusive charged-particle jet cross-section measurement at  $\sqrt{s} = 13.6$  TeV (preliminary; paper in preparation; EPS-HEP 2025 PWG-JE merge talk on behalf of ALICE).  
Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.*

## Education

**Ph.D. Candidate, Physics**, Sungkyunkwan University 2022.03 – expected 2027.02  
Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at  $\sqrt{s} = 13.6$  TeV with ALICE*.  
Defense expected 2026.12; degree conferral 2027.02.  
**B.S., Physics**, Sungkyunkwan University 2021.02

## Appointments and Collaboration Membership

ALICE Collaboration, CERN 2022.10 – present  
ePIC Collaboration, Brookhaven National Laboratory 2024.02 – present  
LAMPS Collaboration, RAON 2022.03 – 2024.04

## Research Highlights

- **Main analyzer**, first ALICE Run 3 inclusive charged-particle jet cross section at  $\sqrt{s} = 13.6$  TeV: responsible for the full chain (O<sup>2</sup>Physics task, QA, unfolding, and a systematic-uncertainty program carried down to the track level); PWG-JE-approved preliminary; presented on behalf of ALICE at EPS-HEP 2025 (PWG-JE merge talk).
- **PWG-JE tracking-efficiency systematics for Hard Probes 2026**: derived the pp charged-particle tracking-efficiency uncertainty at  $\sqrt{s} = 13.6$  and 5.36 TeV ( $p_T$ -binned, 2022/2023/2024 periods), combining the track-selection and ITS-TPC matching components.
- **Central-framework contribution to ALICE O<sup>2</sup>Physics** (AliceO2Group/O2Physics, 2023 – present): authored the PWG-JE cross-section normalization framework used by the SKKU Run 3 charged-jet analyses (pp 13.6 TeV and OO – the latter led by two MSc co-analyzers I mentor), and merged a central-framework fix to the track momentum-resolution smearing methodology shared across analyses.
- **ePIC Barrel Imaging Calorimeter**: lead analyst for the CERN PS T10 (2025.07) Pb/scintillating-fiber campaign; DAQ and prompt analysis at KEK AF-AR (2025.03); co-author on the 2024.08 campaign paper (submitted, NIMA-D-26-00482; arXiv:2604.22647), with module QA and assembly.

## Selected Publications

INSPIRE: [inspirehep.net/authors/2117232](https://inspirehep.net/authors/2117232); ALICE Collaboration co-author since 2022.10; ECCE/ePIC and LAMPS collaboration papers are included in the INSPIRE record.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* (submission expected 2026).  
[main analyzer]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the BIC at the EIC,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; arXiv:2604.22647).  
[co-author; module support and PMT/module performance characterization]
- Y. J. Kim et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682.  
[co-author; Time-of-Flight maintenance and on-site checks]

## Invited and Contributed Talks (since 2022)

**Summary**: 3 invited workshops · 5 invited collaboration-meeting talks · 8 international conference contributions (including the EPS-HEP 2025 merge talk on behalf of ALICE and a Hard Probes 2026 parallel talk) · 9 domestic (KPS) · regular PWG-JE working-group speaker since 2023.

### Invited (external workshops, seminars).

- **FKPPN Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **HIM 2025**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05

- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

#### Invited collaboration-meeting talks.

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3  $pp$  collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet  $O^2$  MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

#### Contributed talks and posters.

- **Hard Probes 2026**, Nashville [international, talk] — *Jet-like dihadron correlations in  $pO$ ,  $OO$ , and  $Ne$ – $Ne$  collisions with ALICE* — 2026.06
- **Hard Probes 2026**, Nashville [international, poster] — *Characterizing inclusive charged-particle jet production and fragmentation in  $pp$  collisions* — 2026.06
- **EPS-HEP 2025**, Marseille [on behalf of the ALICE Collaboration; PWG-JE merge talk] — *Inclusive and semi-inclusive jet cross-section measurements in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.07
- **INPC 2025**, Daejeon [international, talk] — *Charged-particle jet and jet quenching measurement in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV during LHC Run 3 with ALICE* — 2025.05
- **Quark Matter 2025**, Frankfurt [international, poster] — *Inclusive charged-particle jet spectrum in  $pp$  collisions in Run 3 at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.04
- **Hard Probes 2024**, Nagasaki [international, poster] — *First measurements of charged-particle jet production in  $pp$  collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2024.09
- **ATHIC 2023**, Incheon [international, poster] — *Dijet studies at the LHC* — 2023.04
- **Korean Physical Society (KPS) Spring and Fall meetings** [domestic] — 9 contributed talks and posters across Spring and Fall meetings, 2022.10 – 2026.04 (every meeting since joining ALICE).

#### Detector and Test-beam Experience

- **ePIC Barrel Imaging Calorimeter** — three beam-test campaigns — **CERN PS T10** (2025.07; lead analyst), **KEK AF-AR** (2025.03; DAQ, equalization, prompt analysis), and **CERN PS T10** (2024.08; module performance characterization); ran the PMT-module QA for the prototype modules and contributed Pb/scintillating-fiber bundle assembly; JANA2 / EIC-recon studies for sampling-fraction and energy response.
- **ALICE ITS2 alignment monitoring** (CTF level) — service work providing period-by-period tracking-performance diagnostics from compressed time-frame data.
- **LAMPS Forward Tracker** (2022 – 2024) — MPPC/scintillator-plane forward-tracking concept design and Geant4 feasibility simulations.

#### Service and Mentoring

- **Paper Committee member**, ALICE PWG-JE —  $OO$  charged-particle jet measurement led by W. Ham (SKKU).
- Mentor of two MSc co-analyzers on the ALICE Run 3  $OO$  charged-jet measurement — W. Ham (on jet reconstruction, unfolding, normalization, systematic uncertainties, and model comparisons) and one further co-mentee; co-author with W. Ham on the corresponding PWG-JE Analysis Note.
- Co-organizer and instructor of the **KoALICE  $O^2$ Physics Tutorial** (SKKU, 2024.01.22–23; ~22 participants from the Korean ALICE community; co-organized with H. Lee).
- Developed and shared the ePIC Korea-BIC beam-test analysis framework across the Korean BIC institutes (SKKU, PNU, KNU).

## Technical Skills

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**Programming.** C++, Python, ROOT, Geant4, JANA2 / EIC-recon, FastJet, Bash, Git,  $\LaTeX$ .

**Software contributions.** ALICE O<sup>2</sup>Physics (AliceO2Group/O2Physics) since 2023 — jet-finding and response-matrix tasks, systematic-uncertainty pipelines, particle-level cross-section normalization, shared track-smearing tools.

**Event generators.** PYTHIA 8 (Monash, CR modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.

**Analysis.** jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

**Detector hardware.** PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

**Languages.** English (professional working proficiency), Korean (native).

## References

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Available on request.